

Question 6: [Implementation of FEM, 4 Points]

Consider the Poisson problem

$$\begin{aligned} -\Delta u &= f & \text{in } \Omega, \\ u &= g & \text{on } \Gamma, \end{aligned}$$

where $\Omega = [0, 2]^2$. The domain is triangulated using a criss-cross mesh with vertices

$(0, 0)$, $(2, 0)$, $(2, 2)$, $(0, 2)$, and the interior node $(1, 1)$.

The triangulation is obtained by connecting the interior node $(1, 1)$ to each of the four corner vertices, thus forming four triangles. Using conforming piecewise linear finite elements ($\mathbb{P}_1$ elements), derive the finite element system of equations

$$\mathbf{A}\mathbf{u} = \mathbf{b}.$$

- Write down the local stiffness matrices for each triangle.
- Assemble the global stiffness matrix $\mathbf{A}$ and the load vector $\mathbf{b}$ corresponding to this triangulation.
- Indicate how Dirichlet boundary conditions at the four corner nodes are incorporated into the system.

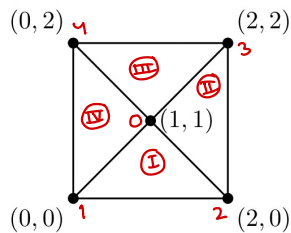
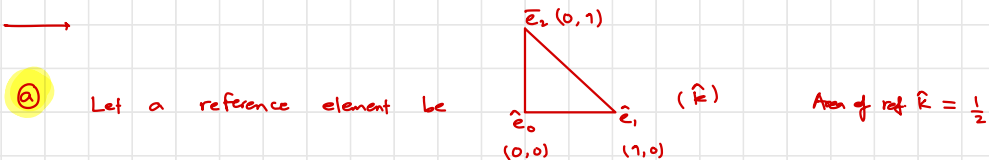


Abbildung 2: Criss-cross triangulation of $\Omega = [0, 2]^2$ for the $\mathbb{P}_1$ finite element.



Then basis functions for reference element of $\mathbb{P}_1$ type is,

$$\hat{\varphi}_0 = 1 - x - y, \quad \hat{\varphi}_1 = x, \quad \hat{\varphi}_2 = y$$

$$\therefore \bar{\mathbf{D}}\hat{\varphi}_0 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \bar{\mathbf{D}}\hat{\varphi}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \bar{\mathbf{D}}\hat{\varphi}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Let A_I local-stiffness matrix formed by vertices $\{0, 1, 2\}$
 A_{II} " " " " " " $\{0, 2, 3\}$
 A_{III} " " " " " " $\{0, 3, 4\}$
 A_{IV} " " " " " " $\{0, 4, 1\}$

A_I : Let a map $F_{K_I} : \{\hat{x}_0, \hat{x}_1, \hat{x}_2\} \rightarrow \{0, 1, 2\}$

$$F_{K_I}(\hat{x}) = B_{K_I} \hat{x} + b_{K_I}$$

where, $B_{K_I} = [\bar{a}_1 - \bar{a}_0 \quad \bar{a}_2 - \bar{a}_0]$, $b_{K_I} = [\bar{a}_0]$

As, $\bar{a}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\bar{a}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\bar{a}_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$$\begin{aligned} \therefore F_{K_I}(\hat{x}) &= \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -\hat{x}_1 + \hat{x}_2 + 1 \\ -\hat{x}_1 - \hat{x}_2 + 1 \end{bmatrix} \end{aligned}$$

So, $B_{K_I} = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \Rightarrow \det(B_{K_I}) = 2$

$$\Rightarrow B_{K_I}^T B_{K_I} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

$$\begin{aligned} (A_I)_{00} &= \int_{K_I} \bar{\psi}_0 \bar{\psi}_0 d\bar{x} = \int_{\hat{K}} (\hat{\psi}_0^T)^T (B_{K_I}^T B_{K_I})^{-1} (\hat{\psi}_0) \det(B_{K_I}) d\hat{x} \\ &= \int_{\hat{K}} [-1 \ -1] \cdot 2I \cdot \begin{bmatrix} -1 \\ -1 \end{bmatrix} \cdot 2 d\hat{x} = 8 \int_{\hat{K}} d\hat{x} = 4 \end{aligned}$$

$$\therefore (A_I)_{00} = 4$$

Similarly,

$$(A_{\mathbf{I}})_{01} = \int_{\hat{\mathbf{K}}} (\hat{\Phi} \hat{\Phi}_0)^T (B_{\mathbf{K}_{\mathbf{I}}}^T B_{\mathbf{K}_{\mathbf{I}}})^{-1} (\hat{\Phi} \varphi_1) \det(B_{\mathbf{K}_{\mathbf{I}}}) d\hat{\mathbf{x}}$$

$$= \int_{\hat{\mathbf{K}}} [-1 \ -1] \begin{bmatrix} 1 \\ 0 \end{bmatrix} 4 d\hat{\mathbf{x}} = -2$$

$$(A_{\mathbf{I}})_{02} = \int_{\hat{\mathbf{K}}} (\hat{\Phi} \hat{\Phi}_0)^T (B_{\mathbf{K}_{\mathbf{I}}}^T B_{\mathbf{K}_{\mathbf{I}}})^{-1} (\hat{\Phi} \varphi_2) \det(B_{\mathbf{K}_{\mathbf{I}}}) d\hat{\mathbf{x}}$$

$$= \int_{\hat{\mathbf{K}}} [-1 \ -1] \begin{bmatrix} 0 \\ 1 \end{bmatrix} 4 d\hat{\mathbf{x}} = -2$$

$$(A_{\mathbf{I}})_{11} = \int_{\hat{\mathbf{K}}} (\hat{\Phi} \hat{\Phi}_1)^T (B_{\mathbf{K}_{\mathbf{I}}}^T B_{\mathbf{K}_{\mathbf{I}}})^{-1} (\hat{\Phi} \varphi_1) \det(B_{\mathbf{K}_{\mathbf{I}}}) d\hat{\mathbf{x}}$$

$$= \int_{\hat{\mathbf{K}}} [1 \ 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} 4 d\hat{\mathbf{x}} = 2$$

$$(A_{\mathbf{I}})_{12} = \int_{\hat{\mathbf{K}}} (\hat{\Phi} \hat{\Phi}_1)^T (B_{\mathbf{K}_{\mathbf{I}}}^T B_{\mathbf{K}_{\mathbf{I}}})^{-1} (\hat{\Phi} \varphi_2) \det(B_{\mathbf{K}_{\mathbf{I}}}) d\hat{\mathbf{x}}$$

$$= \int_{\hat{\mathbf{K}}} [1 \ 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} 4 d\hat{\mathbf{x}} = 0$$

$$(A_{\mathbf{I}})_{22} = \int_{\hat{\mathbf{K}}} (\hat{\Phi} \hat{\Phi}_2)^T (B_{\mathbf{K}_{\mathbf{I}}}^T B_{\mathbf{K}_{\mathbf{I}}})^{-1} (\hat{\Phi} \varphi_2) \det(B_{\mathbf{K}_{\mathbf{I}}}) d\hat{\mathbf{x}} = \int_{\hat{\mathbf{K}}} \begin{matrix} 0 & 1 & 2 \\ [0 \ 1] \end{matrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} 4 d\hat{\mathbf{x}} = 2$$

As, $A_{\mathbf{I}}$ is symmetric, $\therefore A_{\mathbf{I}} = \begin{bmatrix} 4 & -2 & -2 \\ -2 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix} \begin{matrix} 0 \\ 1 \\ 2 \end{matrix}$

$$\underline{\underline{A_{II}}} : \text{ Let } F_{K_{II}} : \{\bar{e}_0, \bar{e}_1, \bar{e}_2\} \longrightarrow \{0, 2, 3\}$$

$$\text{Then, } B_{K_{II}} = [\bar{a}_1 - \bar{a}_0 \quad \bar{a}_3 - \bar{a}_0] \quad , \quad \bar{a}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad , \quad \bar{a}_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad , \quad \bar{a}_3 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$\therefore \det(B_{K_{II}}) = 2 \quad , \quad B_{K_{II}}^T B_{K_{II}} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

$$\therefore (A_{II})_{00} = \int_{\hat{K}} (\hat{\Phi} \hat{\Phi}_0)^T (B_{K_{II}}^T B_{K_{II}})^{-1} (\hat{\Phi} \varphi_0) \det(B_{K_{II}}) d\hat{x}$$

$$= \int_{\hat{K}} \begin{bmatrix} -1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} 4 d\hat{x} = 4$$

$$(A_{II})_{02} = \int_{\hat{K}} (\hat{\Phi} \hat{\Phi}_0)^T (B_{K_{II}}^T B_{K_{II}})^{-1} (\hat{\Phi} \varphi_2) \det(B_{K_{II}}) d\hat{x}$$

$$= \int_{\hat{K}} \begin{bmatrix} -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} 4 d\hat{x} = -2$$

$$(A_{II})_{03} = \int_{\hat{K}} (\hat{\Phi} \hat{\Phi}_0)^T (B_{K_{II}}^T B_{K_{II}})^{-1} (\hat{\Phi} \varphi_3) \det(B_{K_{II}}) d\hat{x}$$

$$= \int_{\hat{K}} \begin{bmatrix} -1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} 4 d\hat{x} = -2$$

$$(A_{II})_{22} = \int_{\hat{K}} (\hat{\Phi} \hat{\Phi}_1)^T (B_{K_{II}}^T B_{K_{II}})^{-1} (\hat{\Phi} \varphi_1) \det(B_{K_{II}}) d\hat{x} = 2$$

$$(A_{II})_{23} = \int_{\hat{K}} (\hat{\Phi} \hat{\Phi}_1)^T (B_{K_{II}}^T B_{K_{II}})^{-1} (\hat{\Phi} \varphi_2) \det(B_{K_{II}}) d\hat{x} = 0$$

$$(A_{II})_{33} = \int_{\hat{K}} (\hat{\Phi} \hat{\Phi}_2)^T (B_{K_{II}}^T B_{K_{II}})^{-1} (\hat{\Phi} \varphi_2) \det(B_{K_{II}}) d\hat{x} = 2$$

$$\therefore A_{II} = \begin{bmatrix} 4 & -2 & -2 \\ -2 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix} \begin{matrix} 0 \\ 2 \\ 3 \end{matrix}$$

$$\underline{A_{III}}: F_{k_{III}}: [\hat{e}_0, \hat{e}_1, \hat{e}_2] \longrightarrow (0, 3, 4)$$

$$B_{k_{III}} = [\bar{a}_3 - \bar{a}_0 \quad \bar{a}_4 - \bar{a}_0] \quad , \quad \bar{a}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \bar{a}_3 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \bar{a}_4 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\therefore \det(B_{k_{III}}) = 2 \quad , \quad B_{k_{III}}^T B_{k_{III}} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = 2I$$

Similarly,

$$A_{III} = \begin{bmatrix} 4 & -2 & -2 \\ -2 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix} \begin{matrix} 0 \\ 3 \\ 4 \end{matrix}$$

$$A_{IV} = \begin{bmatrix} 4 & -2 & -2 \\ -2 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix} \begin{matrix} 0 \\ 4 \\ 1 \end{matrix}$$

⑥ Global Stiffness Matrix : A , b = load vector

$$A_{5 \times 5} U_{5 \times 1} = b_{5 \times 1}$$

$$\text{where } U = \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}_{5 \times 1}, \quad b = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}_{5 \times 1}$$

Start with $A = 0_{5 \times 5}$, $b = 0_{5 \times 1}$

for cell ① : $\{0, 1, 2\}$

$$A_{00} + = (A_I)_{00} = 4, \quad A_{01} + = (A_I)_{01} = -2, \quad A_{02} + = (A_I)_{02} = -2$$

$$A_{10} + = -2, \quad A_{11} + = (A_I)_{11} = 2, \quad A_{12} + = (A_I)_{12} = 0$$

$$A_{20} + = -2, \quad A_{21} + = 0, \quad A_{22} + = (A_I)_{22} = 2$$

for cell ② : $\{0, 2, 3\}$

$$A_{00} + = (A_{II})_{00} = 4, \quad A_{02} + = (A_{II})_{02} = -2, \quad A_{03} + = (A_{II})_{03} = -2$$

$$A_{20} + = -2, \quad A_{22} + = (A_{II})_{22} = 2, \quad A_{23} + = (A_{II})_{23} = 0$$

$$A_{30} + = -2, \quad A_{32} + = 0, \quad A_{33} + = (A_{II})_{33} = 2$$

for cell ③ : $\{0, 3, 4\}$

$$A_{00} + = (A_{III})_{00} = 4, \quad A_{03} + = (A_{III})_{03} = -2, \quad A_{04} + = (A_{III})_{04} = -2$$

$$A_{30} + = -2, \quad A_{33} + = (A_{III})_{33} = 2, \quad A_{34} + = (A_{III})_{34} = 0$$

$$A_{40} + = -2, \quad A_{43} + = 0, \quad A_{44} + = (A_{III})_{44} = 2$$

for cell ④ : $\{0, 4, 1\}$

$$A_{00} + = (A_{IV})_{00} = 4, \quad A_{04} + = (A_{IV})_{04} = -2, \quad A_{01} + = (A_{IV})_{01} = -2$$

$$A_{40} + = -2, \quad A_{44} + = (A_{IV})_{44} = 2, \quad A_{41} + = (A_{IV})_{41} = 0$$

$$A_{10} + = -2, \quad A_{14} + = 0, \quad A_{11} + = (A_{IV})_{11} = 2$$

Finally,

$$\Rightarrow A = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 16 & -4 & -4 & -4 & -4 \\ -4 & 4 & 0 & 0 & 0 \\ -4 & 0 & 4 & 0 & 0 \\ -4 & 0 & 0 & 4 & 0 \\ -4 & 0 & 0 & 0 & 4 \end{bmatrix} \end{matrix}$$

Ans

As,

$$b_i = \sum_K \int_K f \varphi_i d\hat{x}$$

$$= \sum_K \int_K f(\hat{x}) \hat{\varphi}_i(\hat{x}) \underbrace{\det(B_{K_i})}_{=2} d\hat{x}$$

Cell ①: $b_0 + = 2 \int_K f(\hat{x}) \hat{\varphi}_0(\hat{x}) d\hat{x} = \alpha$
 $\{0,1,2\}$

$b_1 + = 2 \int_K f(\hat{x}) \hat{\varphi}_1(\hat{x}) d\hat{x} = \beta$

$b_2 + = 2 \int_K f(\hat{x}) \hat{\varphi}_2(\hat{x}) d\hat{x} = \gamma$

Cell ②: $b_0 + = 2 \int_K f(\hat{x}) \hat{\varphi}_0(\hat{x}) d\hat{x}$
 $\{0,2,3\}$

$b_2 + = 2 \int_K f(\hat{x}) \hat{\varphi}_2(\hat{x}) d\hat{x}$

$b_3 + = 2 \int_K f(\hat{x}) \hat{\varphi}_3(\hat{x}) d\hat{x}$

Cell ③: $b_0 + = 2 \int_K f(\hat{x}) \hat{\varphi}_0(\hat{x}) d\hat{x}$
 $\{0,3,4\}$

$b_3 + = 2 \int_K f(\hat{x}) \hat{\varphi}_3(\hat{x}) d\hat{x}$

$b_4 + = 2 \int_K f(\hat{x}) \hat{\varphi}_4(\hat{x}) d\hat{x}$

Cell ④: $b_0 + = 2 \int_K f(\hat{x}) \hat{\varphi}_0(\hat{x}) d\hat{x}$
 $\{0,1,1\}$

$b_1 + = 2 \int_K f(\hat{x}) \hat{\varphi}_1(\hat{x}) d\hat{x}$

$b_1 + = 2 \int_K f(\hat{x}) \hat{\varphi}_2(\hat{x}) d\hat{x}$

Then, $\Rightarrow b = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} 4\alpha \\ \beta + \gamma \\ \beta + \gamma \\ \beta + \gamma \\ \beta + \gamma \end{bmatrix}$

where α, β & γ integral can be approximated by Numerical quadrature.

Ans

Partitioning Global matrix A into Interior & Boundary nodes

Interior : 0
Node

Boundary : 1, 2, 3, 4
nodes

$$\therefore A = \begin{bmatrix} A_{00} & A_{01} & A_{02} & A_{03} & A_{04} \\ A_{10} & A_{11} & A_{12} & A_{13} & A_{14} \\ A_{20} & A_{21} & A_{22} & A_{23} & A_{24} \\ A_{30} & A_{31} & A_{32} & A_{33} & A_{34} \\ A_{40} & A_{41} & A_{42} & A_{43} & A_{44} \end{bmatrix} = \begin{bmatrix} A_{in} & B \\ B^T & A_{bc} \end{bmatrix}$$

where $A_{in} = [A_{00}] \equiv$ correspond to inner nodes

$B = [A_{01} \ A_{02} \ A_{03} \ A_{04}] \equiv$ correspond to coupling b/w
Inner & boundary nodes

$A_{bc} = \begin{bmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{bmatrix} \equiv$ correspond to boundary nodes

As, Poisson eqⁿ translated to FE as,

$$A U = b$$

$$\text{where } U = \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} U_{in} \\ U_{bc} \end{bmatrix} \Rightarrow \begin{matrix} \bar{U}_{in} = [u_0] \\ \bar{U}_{bc} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} \end{matrix}$$

$$b = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} b_{in} \\ b_{bc} \end{bmatrix} \Rightarrow \begin{matrix} \bar{b}_{in} = b_0 \\ \bar{b}_{bc} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \end{matrix}$$

$$\therefore \begin{bmatrix} A_{in} & B \\ B^T & A_{bc} \end{bmatrix} \begin{bmatrix} u_i \\ u_{bc} \end{bmatrix} = \begin{bmatrix} b_{in} \\ b_{bc} \end{bmatrix}$$

Now to impose Dirichlet BC, we need to set

$$A_{bc} = I_{4 \times 4}$$

$$\Delta \quad b_{bc} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{bmatrix} = g$$

$$\therefore A_{in} u_{in} + B u_{bc} = b_{in} \Rightarrow A_{00} u_0 + B u_{bc} = b_0$$

$$\Delta \quad B^T u_{in} + I \cdot u_{bc} = g \Rightarrow B^T u_0 + u_{bc} = g$$

$$\therefore u_{bc} = g - B^T u_0$$

$$\text{Then} \quad A_{00} u_0 + B (g - B^T u_0) = b_0$$

$$(A_{00} - B B^T) u_0 = b_0 - B g$$

$$\Rightarrow \left\{ u_0 = \frac{1}{(A_{00} - \underbrace{B B^T}_{\text{number}})} (b_0 - B g) \right\}$$

Ans